

Effect of Geometry on Apparent Stiffness of Triply Periodic Minimal Surface (TPMS) Porous Scaffolds for Orthopaedics Implants



zhewei-k.github.io

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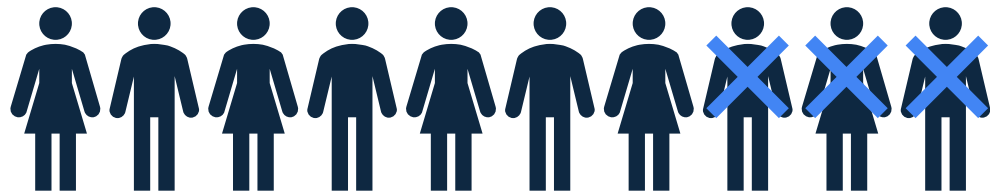
School of
Mechanical, Aerospace
& Civil Engineering

Background | The Problem.

Low Implant Lifespans

Three out of every ten patients face complications in the 25 years after surgery in the UK ^{[1] [2]}.

3-5X higher risk in younger patients ^[3].



High Costs

Total hip and knee replacements cost **£2 billion** ^[4] of the annual NHS budget.

+40% increase in operations over the next 30 years ^[5].

+40% in number of operations

[1] J. T. Evans, et al. How long does a knee replacement last? A systematic review and meta-analysis of case series and national registry reports with more than 15 years of follow-up, vol. 393, Elsevier BV, 2019, pp. 655-663. [2] J. T. Evans, et al. How long does a hip replacement last? A systematic review and meta-analysis of case series and national registry reports with more than 15 years of follow-up, vol. 393, NLM (Medline), 2019, pp. 647-654. [3] L. E. Bayliss et al, "The effect of patient age at intervention on risk of implant revision after total replacement of the hip or knee: a population-based cohort study," The Lancet, vol. 389, no. 10077, pp. 1424-1430, 4 2017. [4] National Institute for Health and Care Excellence (NICE), "Impact on NHS workforce and resources | Joint replacement (primary): hip, knee and shoulder | Guidance | NICE," 6 2020. [5] G. S. Matharu et al, "Projections for primary hip and knee replacement surgery up to the year 2060: an analysis based on data from The National Joint Registry for England, Wales, Northern Ireland and the Isle of Man," Annals of the Royal College of Surgeons of England, vol. 104, no. 6, pp. 443-448, 6 2022.

Background | The Problem.



Aseptic Loosening

- Long term issue for implants that use bone cement for fixation (standard).
- Caused by stress shielding – mismatch of stiffness between bone and metal implant.

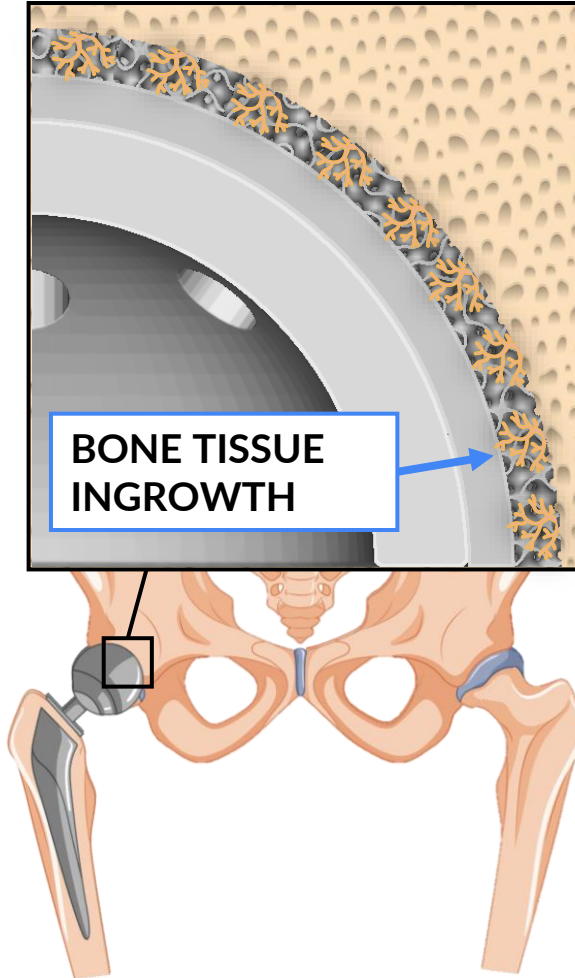
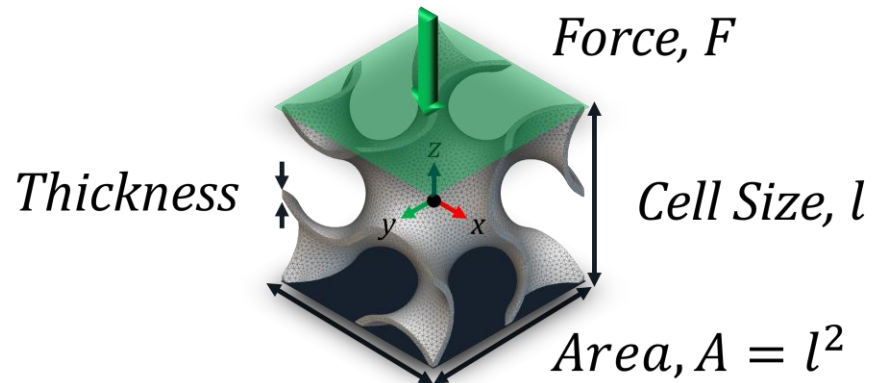
Cementless Implants

Porosity (or relative density) not designed to patient needs.

Background | The Solution.

Porous surfaces with **customised apparent stiffness** using **sheet-TPMS** geometry.

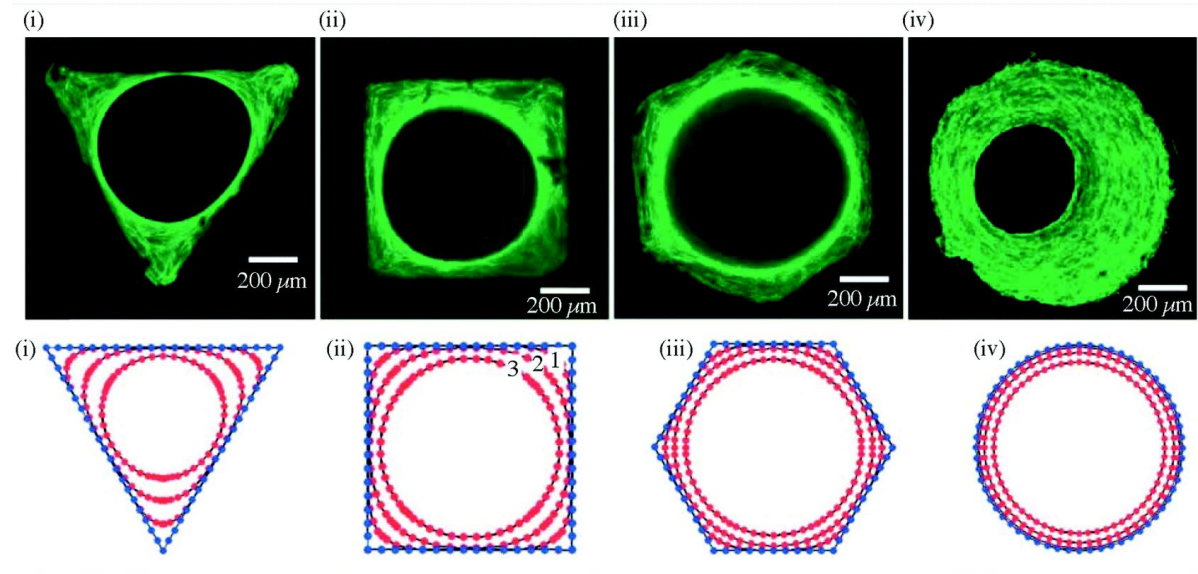
$$E_{\text{scaffold}} = \frac{\sigma}{\epsilon} = \frac{Fl}{A \cdot \Delta l}$$



Background | Why Sheet-TPMS?

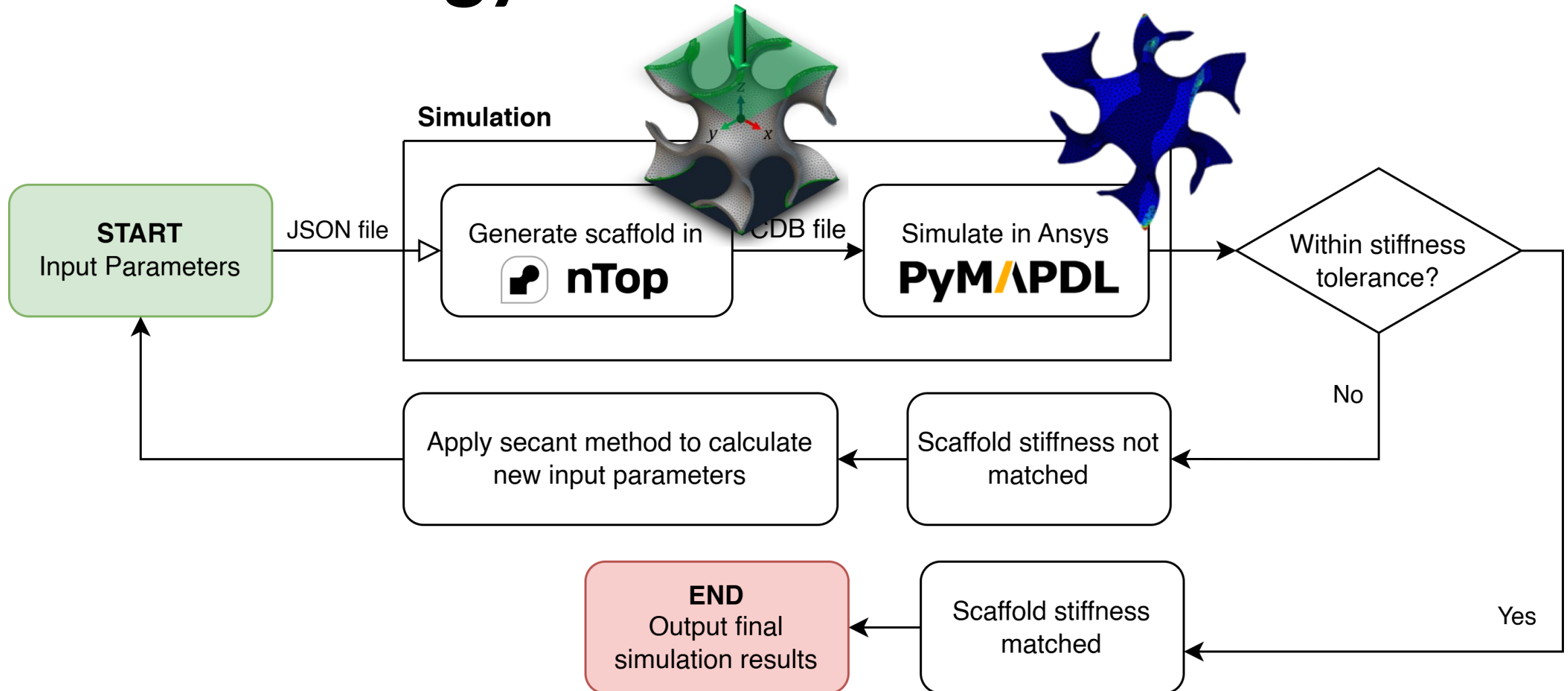
Sheet-TPMS Geometry

- Best environment for **cell proliferation and adhesion** [1].
- **High permeability** offers high oxygenation and nutrition delivery [2].
- **High fatigue resistance** due to smooth contours [3].



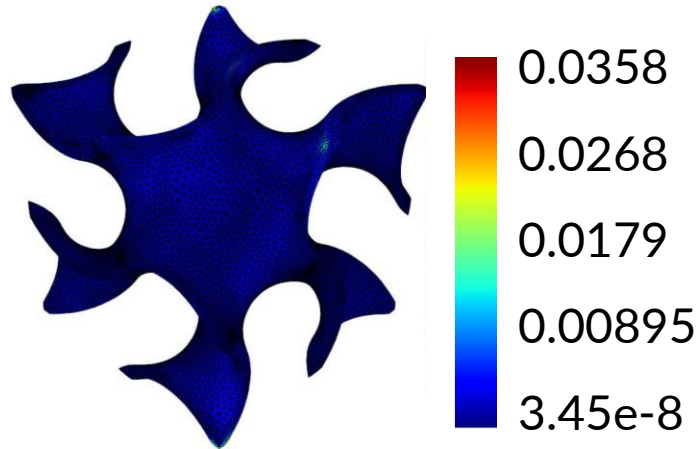
[1] A. A. Zadpoor, "Bone tissue regeneration: the role of scaffold geometry," *Biomaterials Science*, vol. 3, no. 2, pp. 231-245, 1 2015. [2] F. Bobbert et al, "Additively manufactured metallic porous biomaterials based on minimal surfaces: A unique combination of topological, mechanical, and mass transport properties," *Acta Biomaterialia*, vol. 53, pp. 572-584, 4 2017. [3] E. Maevskaia et al, "Triply Periodic Minimal Surface-Based Scaffolds for Bone Tissue Engineering: A Mechanical, In Vitro and In Vivo Study," *Tissue Engineering - Part A*, vol. 29, no. 19-20, pp. 507-517, 10 2023.

Methodology.



Results.

Gyroid (G)

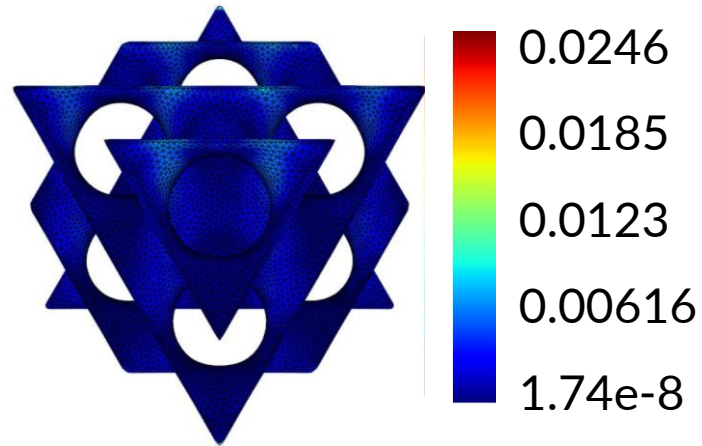


Thickness = 0.145 mm

Porosity = 85.2%

$E_{scaffold} = 2.99$ GPa

Diamond (D)

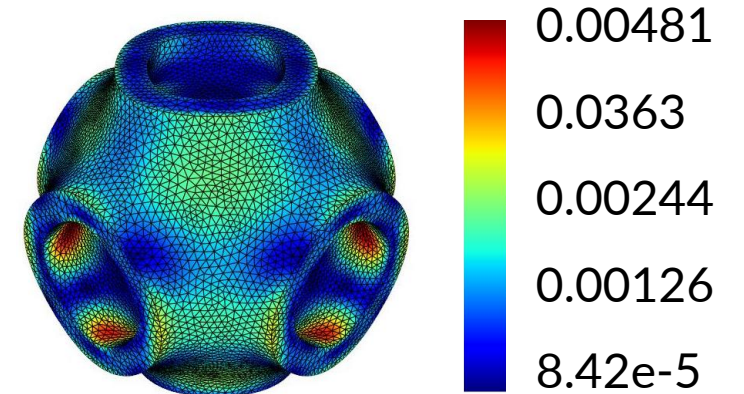


Thickness = 0.0653 mm

Porosity = 91.7%

$E_{scaffold} = 3.01$ GPa

Primitive (P)



Thickness = 0.278 mm

Porosity = 78.4%

$E_{scaffold} = 3.03$ GPa

Results.

Input Parameters

Thickness = 0.1 mm

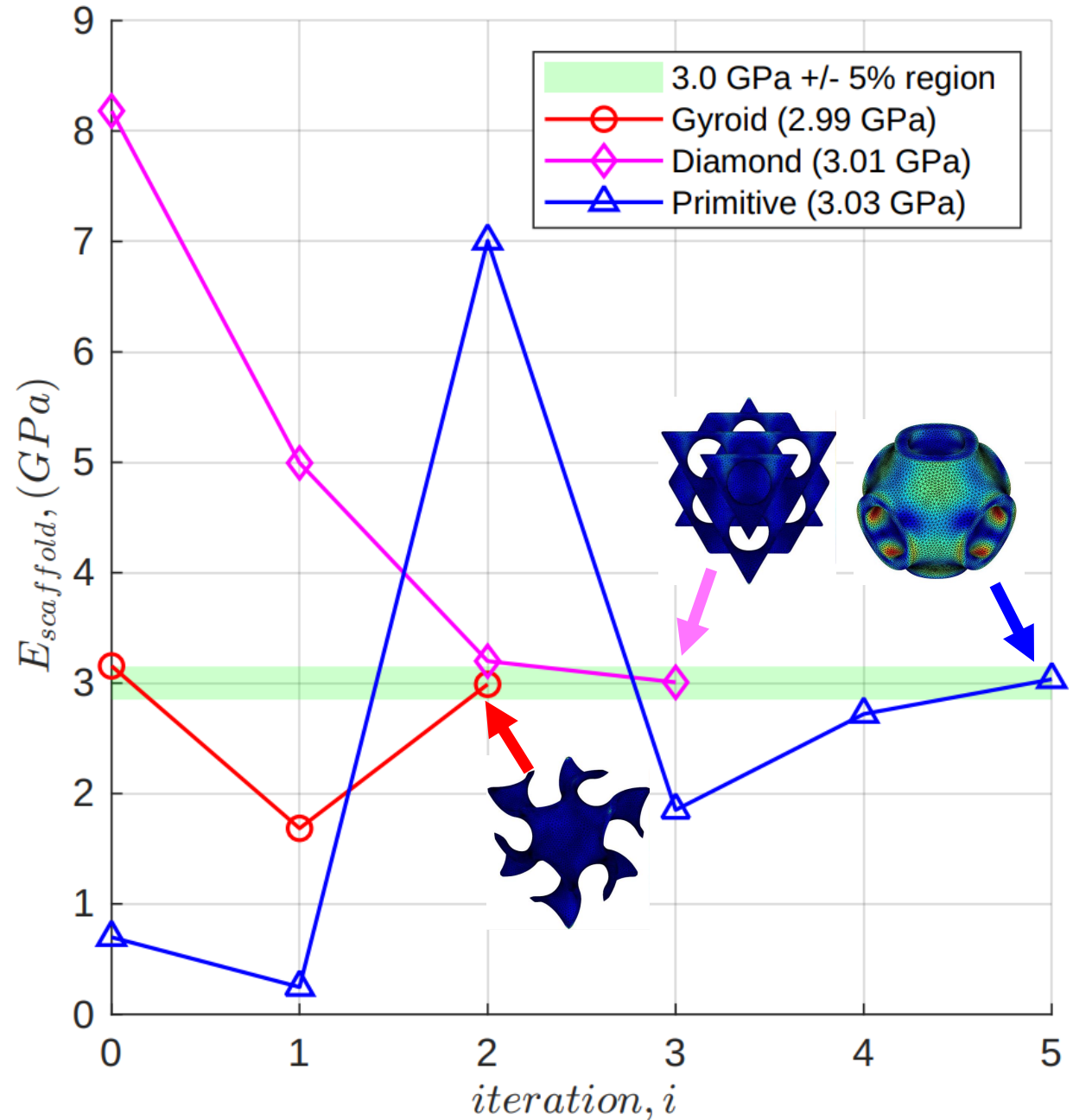
Cell Size = 3 mm

Desired stiffness = 3.0 GPa

Tolerance = $\pm 5\%$

Output

Type	$E_{scaffold}$ (GPa)	% Error
Gyroid	2.99	-0.33%
Diamond	3.01	0.33%
Primitive	3.03	1.00%

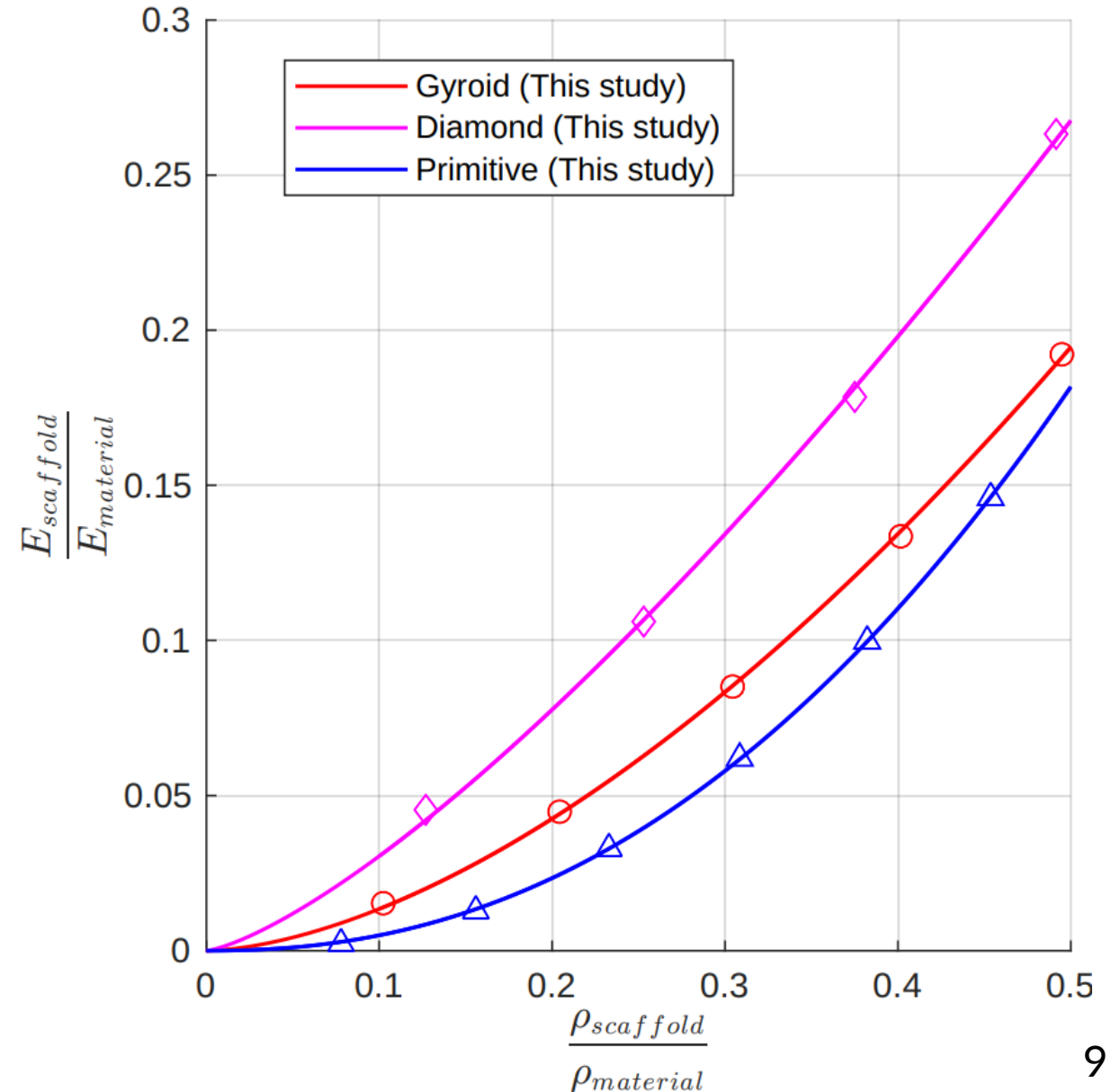


Validation.

Gibson-Ashby Model

Used to interpolate data points and derive analytical solution.

$$\frac{E_{scaffold}}{E_{material}} = C \times \left(\frac{\rho_{scaffold}}{\rho_{material}} \right)^n$$



Validation.

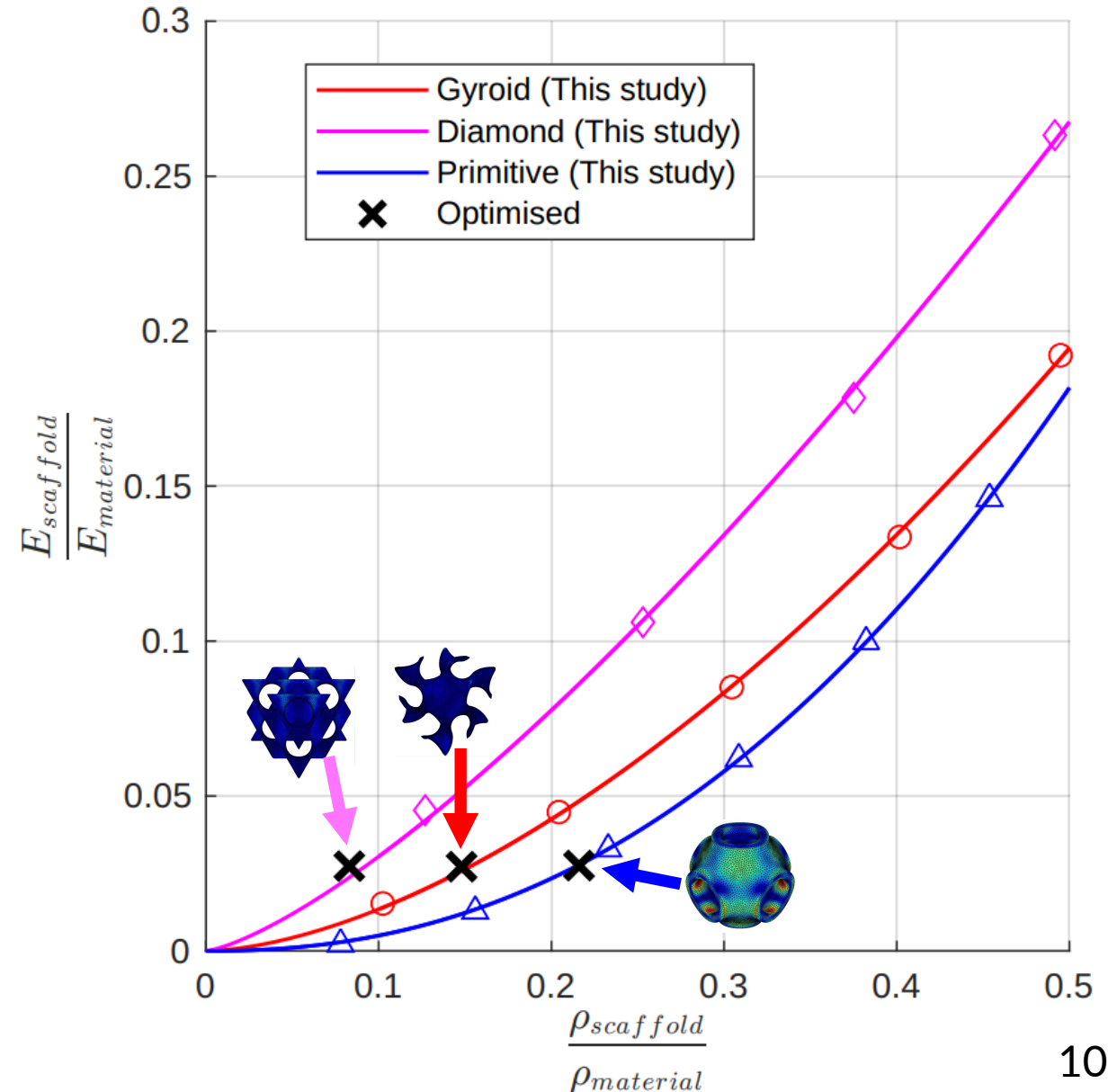
Gibson-Ashby Model

Used to interpolate data points and derive analytical solution.

$$\frac{E_{scaffold}}{E_{material}} = C \times \left(\frac{\rho_{scaffold}}{\rho_{material}} \right)^n$$

Type	Analytical	Numerical (×)	% Error
Gyroid	0.0256	0.0272	6.25%
Diamond	0.0236	0.0274	15.85%
Primitive	0.0278	0.0275	-0.83%

[1] U. Simsek et al, "Modal characterization of additively manufactured TPMS structures: comparison between different modeling methods," International Journal of Advanced Manufacturing Technology, vol. 115, no. 3, pp. 657-674, 7 2021.



Validation.

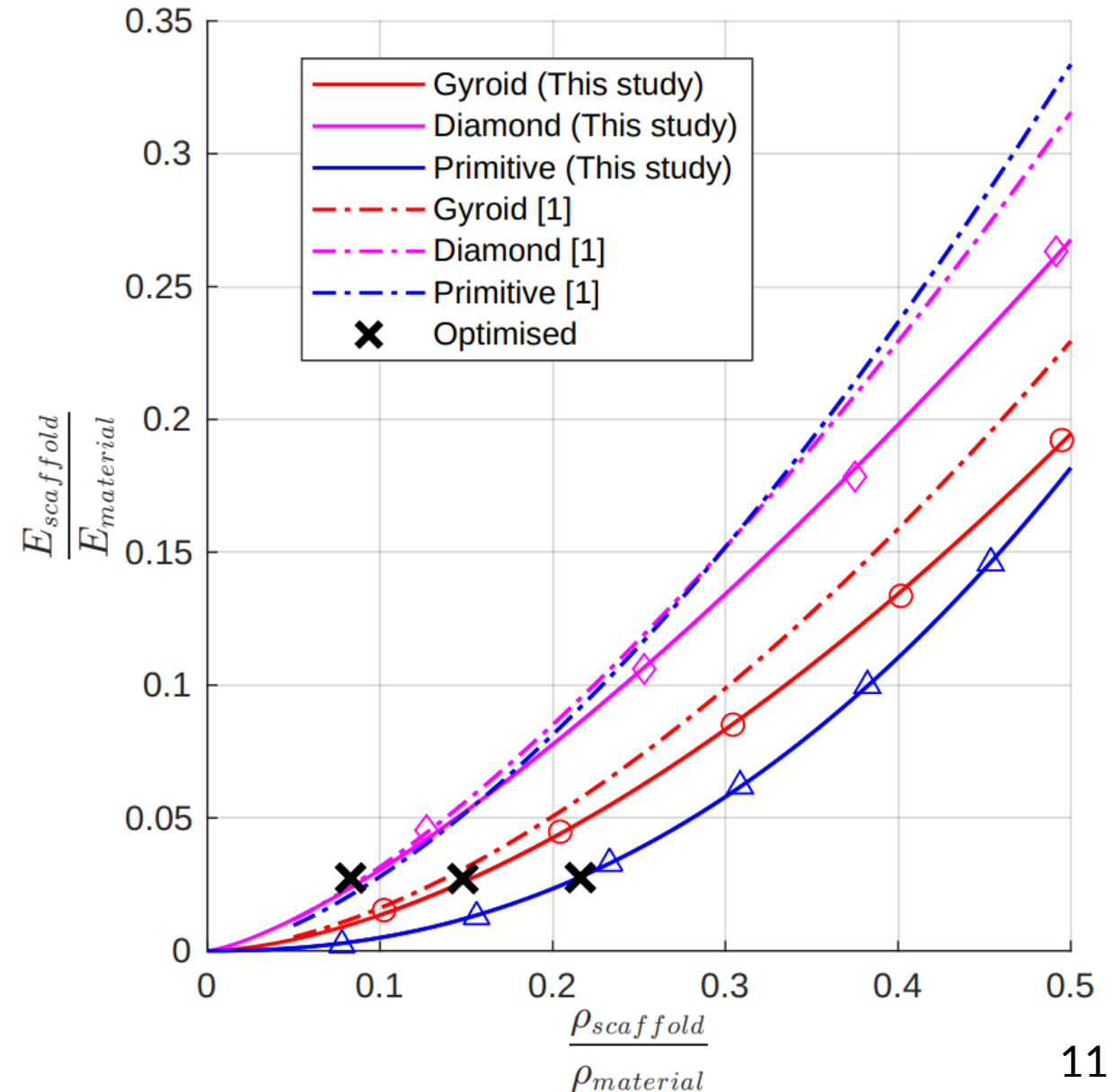
Gibson-Ashby Model

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Conclusion.

Three types of sheet-TPMS scaffolds with a stiffness of 3.0 GPa were developed, with high accuracy.

Limitations & Further Development

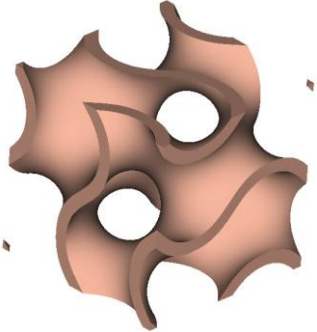
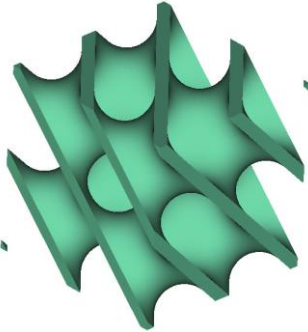
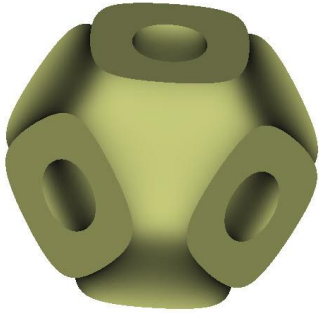
- Lack of consideration for real world scenario.
 - Small thickness ($\sim 0.1\text{mm}$) can result in manufacturing difficulties
 - Research has shown that process parameters play a huge role as well, and need to be considered [1] [2] [3] [4]
- Only 1x1x1 cell configuration has been analysed here.

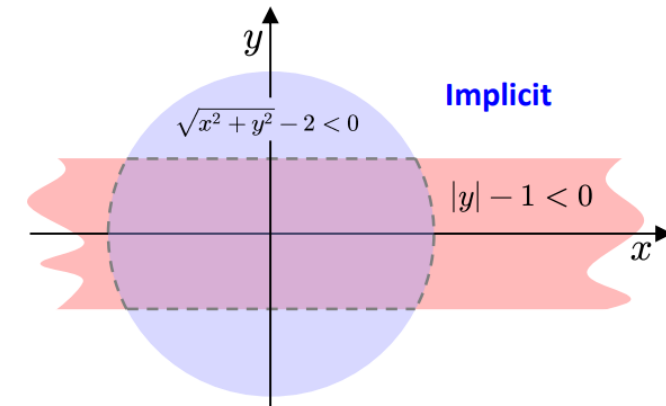
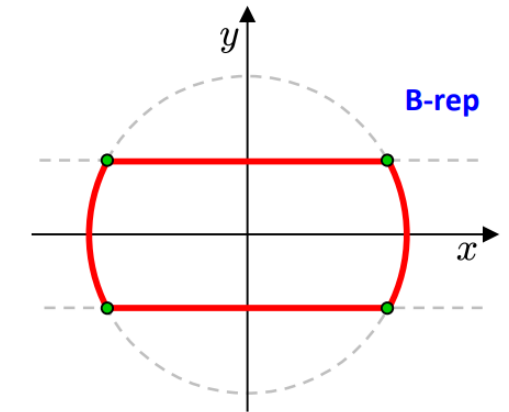
Thank you!

to my supervisor, Vee,
to my personal tutor, Marco,
to George Wilkinson for his
technical support,
and you for listening.



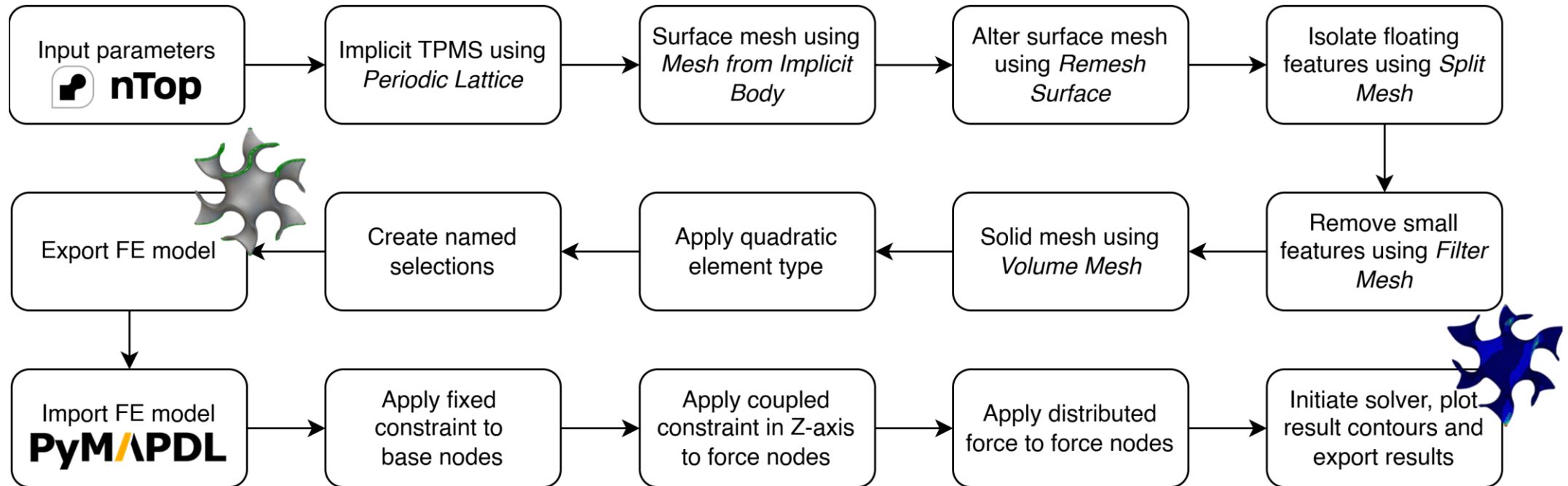
Appendix | Modelling.

Geometry Type	Gyroid (G)	Diamond (D)	Primitive (P)
Image			
Equation	$\sin(x) \sin(y) + \sin(y) \sin(z) + \sin(z) \cos(y)$	$\sin(x) \sin(y) \sin(z) + \sin(x) \cos(y) \cos(z) + \cos(x) \sin(y) \cos(z) + \cos(x) \cos(y) \sin(z)$	$\cos(x) + \cos(y) + \cos(z)$

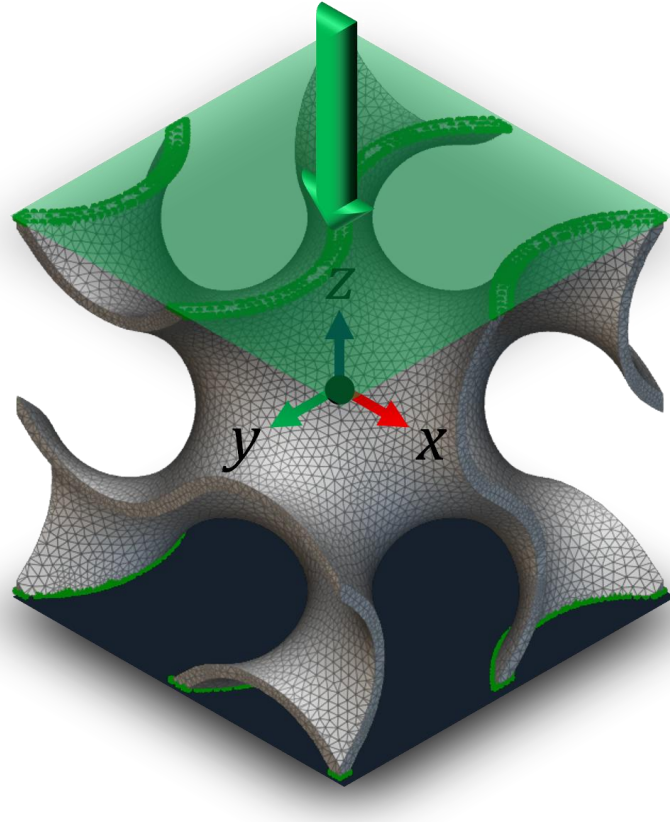


Appendix | Simulation.

Simulation



Appendix | Boundary Conditions.



Force Nodes

$$FZ = -100 \text{ N}$$

$$UZ = \text{constant}$$

$$UX, UY = \text{free}$$

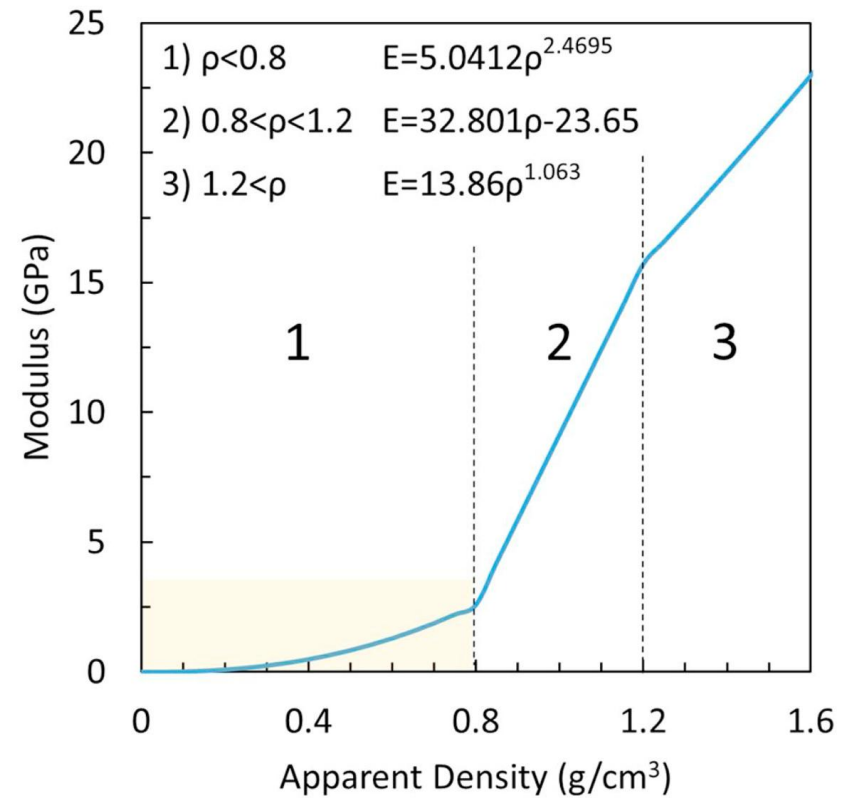
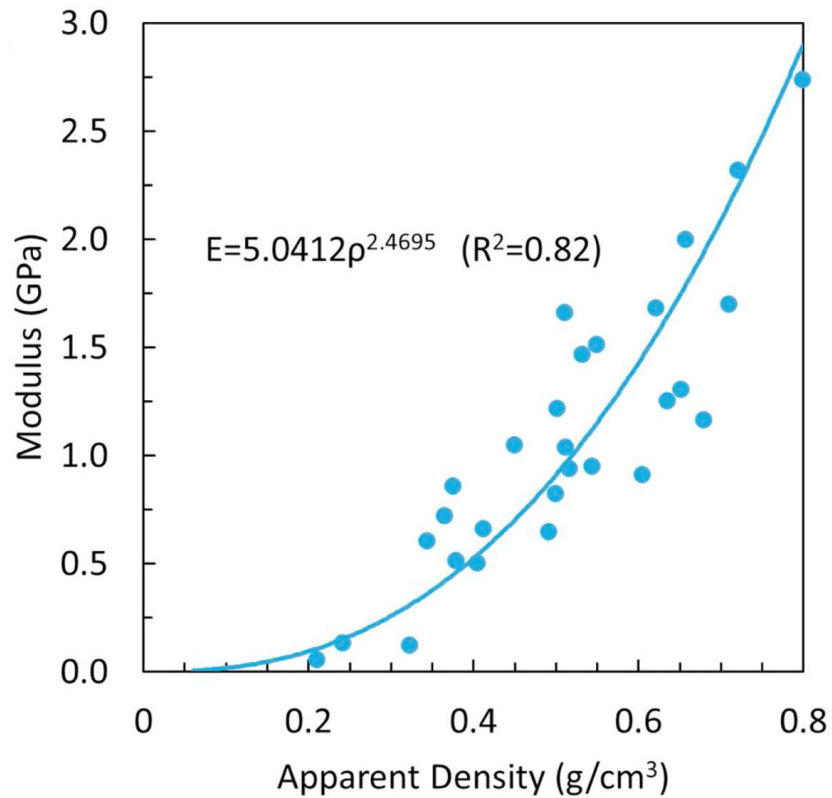
$$RX, RY, RZ = \text{free}$$

Base Nodes

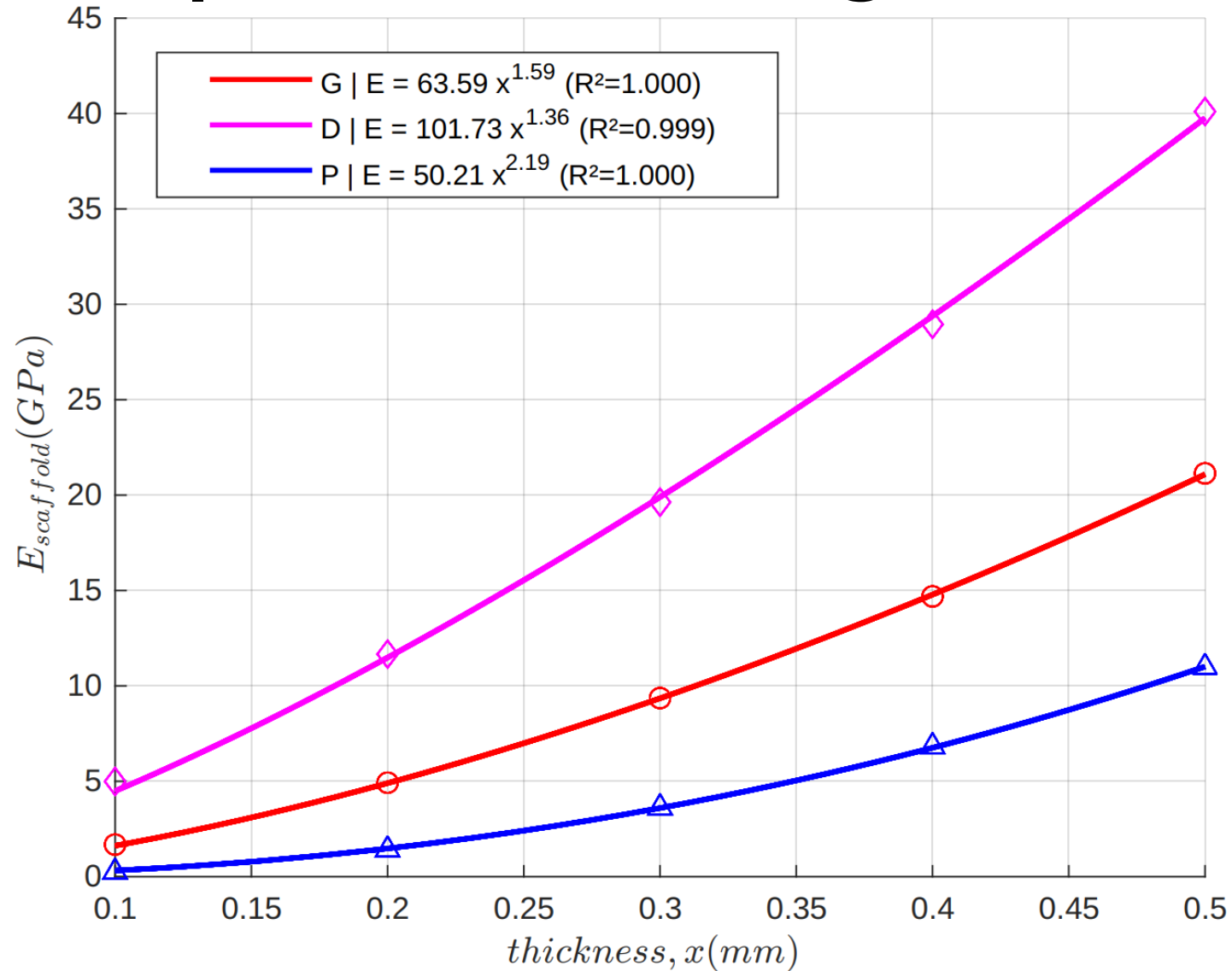
$$UX, UY, UZ = 0$$

$$RX, RY, RZ = 0$$

Appendix | Why 3.0 GPa?



Appendix | Mesh Convergence



Appendix | Gibson-Ashby Equations

